

TNP HOSPITALITY PLATFORM

Detailed Engineering Scope — Phase 1

Internal build specification for the delivery team and AI coding agents

Audience	Shuraya Labs delivery team · Claude Opus 5 · GPT-6 Astra · GPT-5.6 Sol / Codex
Status	Binding. This document defines behaviour; the client proposal defines commitments.
Delivery date	13 October 2026
Scope basis	Merged from the Shuraya Labs scope and the competing TechVerito scope of work
Rule	Where this document and any prompt disagree, this document wins. Do not reinterpret business rules.

1. How This Document Is Used

This is the specification an engineer or an AI agent reads before writing code. It is deliberately more detailed than the client proposal, and it is the source of truth for behaviour.

- Section 3 defines the domain objects. Field names here are the field names in code.
- Section 4 defines the state machines. Do not invent states.
- Section 5 defines business rules that must not be reinterpreted under any circumstances.
- Section 6 defines every module, with acceptance criteria.
- Section 9 lists the traps — the places where generated code is reliably wrong on this project.

An agent that is unsure must stop and ask. A thirty-second question is cheaper than three hours of confidently wrong work that another developer then builds on.

2. System Overview

2.1 Actors

Actor	Authenticated	Description
Visitor	No	Browses the public site, submits enquiries
Client (B2C)	Yes	Individual booking an event. Views quotations, pays, tracks the event
Planner (B2B)	Optional	Event business sourcing manpower. May submit without an account or hold a full portal
Freelancer	Yes	Gig worker. Claims slots, checks in, earns, gets paid monthly
Team Leader	Yes	A Coordinator or Executive designated TL for one specific event. Not a separate account type
Admin	Yes	Six scoped roles — see 6.11

2.2 Technology

Layer	Choice	Note
Web	Next.js / React	App Router. Server components where they help
Styling	Tailwind CSS + shadcn/ui	One shared design system package
Motion	Framer Motion	Restrained. Public site only
Backend	Node.js + Express	API-first. Phase 2 mobile consumes the same API
Database	MongoDB Atlas	BINDING. Not Postgres. The cost model depends on this
Object storage	Cloudflare R2	S3-compatible. No per-GB floor, no egress charge
Cloud	DigitalOcean	Bangalore region for latency
CDN / DNS / SSL	Cloudflare	Free plan
Push	Firebase Cloud Messaging	Primary notification channel. Cheaper than SMS
OTP	SMS provider	Signup verification only. DLT registration required
Payments in	Payment gateway	Client and planner collections
Payouts out	Payout API	Monthly batch to freelancers
Messaging	WhatsApp Business API	RSVP guest workflow. Utility templates only
KYC	Verification API	Freelancer identity verification at application



MongoDB is a binding architectural decision. An agent proposing Postgres, Prisma or a different ORM must write a proposal and stop — it must not implement it. The entire infrastructure cost model quoted to the client assumes MongoDB Atlas.

2.3 Repository Layout

apps/web	Next.js – public site + all four portals
apps/api	Express – modular, one folder per domain module
packages/types	Shared TypeScript interfaces. FROZEN after day 3
packages/ui	Design system. Built once, imported everywhere
docs/	This document, API contract, decisions, task contracts



3. Domain Model

Field names below are the field names in code. Do not rename, do not abbreviate, do not add fields not listed without agreement.

3.1 Core Entities

Entity	Key fields	Notes
User	_id, email, phone, passwordHash, role, status, createdAt	One user table. `role` discriminates client / planner / freelancer / admin
FreelancerProfile	userId, roleLevel, experienceYears, skills[], assessmentScore, kycStatus, payRateOverride, rating, strikeCount, status	roleLevel ∈ coordinator executive volunteer hostess
PlannerProfile	userId, companyName, gstNumber, photo, portfolioText, verifiedAt, accessCode, status	portfolioText and photo required to appear in the B2C planner picker
ClientProfile	userId, name, phone, address	Minimal
Venue	_id, name, location{city,area,coords}, capacity, priceBand, venueType, amenities[], photos[], isActive	Admin-managed. Powers the B2C venue step
Booking	_id, clientId, plannerId, venueId, eventType, guestCount, status	Parent record. A wedding is one Booking
Event	_id, bookingId, title, functionType, venueId, startAt, endAt, teamLeaderId, status	A function. A wedding has several Events under one Booking
Position	_id, eventId, roleLevel, requiredQty, filledQty, payRate	NEVER hard-code 10+1. Quantity is per position, per event
Assignment	_id, positionId, eventId, freelancerId, claimedAt, confirmedAt, status	claimedAt is the FCFS ordering key
Attendance	_id, assignmentId, checkInAt, checkOutAt, checkInCoords, distanceMeters, gpsAvailable, verifiedBy	Append-only
Rating	_id, assignmentId, freelancerId, ratedBy, punctuality, professionalism, communication, workQuality, overall, comment	TL submits within 1–2 days
Earning	_id, freelancerId, assignmentId, amountPaise, adjustmentPaise, adjustmentReason, status	Derived from Attendance only
PayoutBatch	_id, month, freelancerId, totalPaise, idempotencyKey, providerTxnId, status	One per freelancer per month
Quotation	_id, bookingId, version, lineItems[], totalPaise, issuingEntity, status	issuingEntity toggles the invoice name
Invoice	_id, quotationId, number, issuingEntity, totalPaise, paymentStatus	Paid / Unpaid / Overdue
Expense	_id, categoryId, eventId, amountPaise, description, spentAt	eventId optional — enables per-event costing
ExpenseCategory	_id, name, isActive	User-creatable. Not a fixed enum
Guest	_id, eventId, name, phone, rsvpStatus, docStatus, docUrl, transportMode, pnr, lastContactedAt	RSVP module
Lead	_id, source, payload, status, assignedTo	Unified queue for Booking and RSVP enquiries
AuditLog	_id, actorId, action, entityType, entityId, before, after, at	Every sensitive action

4. State Machines

These are exhaustive. An agent must never introduce a state outside these lists, and must never skip a transition.

4.1 Event

draft → staffing → confirmed → in_progress → completed → settled

- `staffing` is entered when Positions are published and freelancers are notified
- `confirmed` requires every Position filledQty == requiredQty and all Assignments confirmed
- `settled` requires all Earnings approved and included in a PayoutBatch

4.2 Assignment

claimed → confirmed → checked_in → checked_out → completed

claimed → released (confirmation deadline passed)

claimed → cancelled (freelancer declines, or admin removes)

- `released` reopens the slot: decrement filledQty in the same atomic operation

4.3 Payout

pending → approved → processing → paid

→ processing → failed → (retry) → processing

- Never write `paid` without a providerTxnId returned by the payout provider
- Never skip a state. `pending → paid` is a defect, not a shortcut

4.4 Guest (RSVP)

pending → contacted → confirmed | not_coming

docStatus: not_required | required | requested | received

4.5 Freelancer account

applied → assessed → kyc_pending → verified → active

active → under_review (three consecutive poor ratings)

under_review → active | suspended | deactivated (ADMIN ACTION ONLY)

5. Business Rules — Binding

These rules are the ones an agent is most likely to quietly reinterpret. They must not be reinterpreted.

5.1 Allocation — first come, first served

- Slots fill by claimedAt, earliest first
- Claiming MUST be a single atomic conditional update. A read followed by a write allows eleven people into ten slots. This is the single most important rule in this document
- A freelancer may hold only one confirmed Assignment per overlapping time window
- Admin override exists: admin may assign manually, bypassing the race, for curated requirements
- Unconfirmed Assignments auto-release at the confirmation deadline and the slot reopens

5.2 Notification and confirmation window

- A push or full-screen notification is sent to each assigned freelancer TWO HOURS before event start
- It requires an explicit Coming / Not Coming response
- No response within one hour is treated as Not Coming; the slot is flagged for replacement
- The two-hour and one-hour values are configurable settings, not constants in code

5.3 Attendance

- Check-in requires a valid QR or code, plus a Team Leader or coordinator action, plus a timestamp
- GPS is validated against venue coordinates within a configurable radius
- GPS may be absent when permission is denied. Record gpsAvailable = false. NEVER block check-in on missing GPS, and NEVER treat absent as valid
- Attendance records are append-only. A correction creates a new record with a reason, it does not edit the original

5.4 Earnings and payouts

- Earnings derive ONLY from verified Attendance. No attendance, no earning. There is no manual earning creation path for freelancers
- Payouts are monthly and consolidated — exactly ONE transfer per freelancer per month, covering all their events that month
- Every payout carries an idempotencyKey. A retried request must never produce a second transfer
- Two approvals required: Team Leader or coordinator, then Finance Admin
- All money is stored as integers in paise. Never floats, anywhere, for any reason

5.5 Ratings

- The Team Leader submits ratings within one to two days of event completion
- Three consecutive poor ratings move the account to `under\_review` and notify admin
- Permanent deactivation is applied by a human admin. Automatic irreversible blocking is behind a feature flag, default OFF
- Admin may promote or demote a freelancer's roleLevel. The applicable pay rate follows the new level from the date of change, not retroactively

5.6 Events and positions

- An Event defines its own Positions and quantities. Ten freelancers plus one coordinator is a DEFAULT, never a hard-coded rule
- A wedding contains multiple functions. Each function is its own Event, grouped under one Booking

5.7 Money and entities



-
- Client collections and freelancer payouts are separate ledgers. Never mix, never net off
 - Invoice issuing entity is selected by a toggle on the Quotation. The entity name on the generated document follows that toggle

5.8 Guest identity documents

- Collect MASKED documents only — last four digits visible. Never store a full identity number
- Stored encrypted in access-controlled object storage, never in the database, never in a spreadsheet
- Visible only to Super Admin and Booking & RSVP Admin
- A retention and deletion policy runs after each event. Implement the deletion job, do not leave it manual

6. Modules

Each module below lists what to build and how it is judged complete.

6.1 Public Website

- Home, About, Services (3 sub-pages), 12 department pages, Portfolio, Join Us, Planners, Book an Event, RSVP Services, Contact
- Department pages generate from a single content collection — do not hand-build twelve templates
- Portfolio filterable by event type, photo and video
- Arch-masked photography treatment, teal and sand palette, scroll-driven horizontal galleries

Done when: every page responsive, Lighthouse performance above 85 on mobile, all imagery from TNP's library, no stock photos.

6.2 Freelancer Onboarding & Assessment

- Application form: personal details, experience, skills, availability, preferred role
- Assessment engine: question set configurable in admin, scored automatically, produces a recommended roleLevel
- KYC via third-party verification API, status stored, failures surfaced to admin
- Admin approval required before portal access. Admin may override the recommended level
- "Track My Application" by reference ID, no login required

Done when: an applicant can complete application → assessment → KYC → approval → portal access end to end, and the assessment question set can be edited in admin without a code change.

6.3 Freelancer Portal

- Opportunity feed — eligible events only, filtered by roleLevel, availability, rating and location
- Claim slot with live "6 of 10 claimed" counter
- Assignment confirmation, Coming / Not Coming response
- Personal QR code for attendance
- Earnings to date, payout history with status and reference
- Ratings received, feedback comments, account standing and strike count

Done when: mobile-first, 500 concurrent claims resolve correctly with no over-assignment, and the feed loads in under one second on a mid-range phone.

6.4 Planner Module

- Open submission form — no account required
- Full planner portal for repeat partners, with five-field registration
- Admin verification and access-code issue
- Requirement creation: roleLevel, quantity, date, time, location, experience
- Status tracking — pending / partially assigned / confirmed — in portal or by reference ID
- Planner profile with photo and portfolio text, feeding the B2C planner picker

Done when: both paths (with and without account) produce the same Requirement record, and a planner without an account can track status by reference ID.

6.5 Client Booking Flow

- Four-step wizard with progress bar: Venue → Planner → Details → Submit
- Venue step: confirm own venue, or provide location and budget for filtered suggestions from the Venue database
- Planner step: TNP Event Planners listed first and marked recommended, then partner planners
- Details step: date, event type, guest count, contact, special requirements

- Client portal: quotation view, approve, pay, event dashboard, assigned team, RSVP summary, documents

Done when: the wizard preserves state across steps and refresh, and venue filtering returns correctly for location plus budget combinations.

6.6 Venue Database

- Admin CRUD, bulk CSV import, deactivate rather than delete
- Fields per 3.1. Photos in object storage
- Indexed on location.city and priceBand

Done when: admin can add a venue and it appears correctly filtered in the client flow within the same session.

6.7 Event & Workforce Management

- Create Event from a Requirement or directly
- Configurable Positions and quantities
- Eligibility filtering, FCFS claiming, admin manual override
- Team Leader designation per Event
- Confirmation deadline, replacement workflow, slot reopening
- Roster, timeline, operational notes, document attachments
- Multi-function support under one Booking

Done when: the concurrency test in 9.1 passes, and an event with a non-standard shape (40 staff, or 6 staff) works without code changes.

6.8 Attendance

- QR generation per freelancer per event
- Team Leader scan, and manual code entry as fallback
- Check-in and check-out with timestamp and coordinates
- Distance validation against venue, configurable radius
- Attendance history per freelancer and per event

Done when: check-in succeeds with GPS denied, and the record correctly shows gpsAvailable false rather than a false pass.

6.9 Ratings & Performance

- Four-criteria rating plus overall and comment, submitted by TL
- Rating history, average, strike tracking
- Three-strike review workflow with admin action
- Promote and demote with pay rate effect from date of change

Done when: three consecutive poor ratings move the account to under_review and nothing is deactivated without an admin action.

6.10 Finance

Earnings and payouts

- Rate resolution: position rate → freelancer override → role default
- Attendance-derived calculation, adjustments with reason
- Two-stage approval, monthly batch, payout API, webhooks, reconciliation

Quotations and invoices

- Quotation builder with line items and versioning
- Client approval recorded, revision requests tracked
- Invoice generated from TNP's supplied template, issuing entity by toggle

- Payment status per invoice, PDF generation, email delivery

Expenses

- Log expense with category, amount, date, description
- Categories user-creatable, not a fixed enum
- Optional eventId tag enabling per-event cost tracking
- Event profitability view: collections minus payouts minus tagged expenses
- Exports by period, category or event

Done when: the idempotency test in 9.2 passes, and an event's profitability view reconciles against its invoices, payouts and expenses.

6.11 Admin Dashboard & RBAC

Role	Scope
Super Admin	Everything, including creating admins and setting permissions
Freelancer Admin	Applications, assessments, KYC, approvals, promotions, pay rates
Planner Admin	Planner verification, profiles, requirement assignment
Booking & RSVP Admin	Leads, RSVP guest workflow, venue database
Finance Admin	Quotations, invoices, payments, earnings, payouts, expenses
Operations Admin	Events, allocation, attendance, ratings, reports

- Executive dashboard: live events, staffing status, pending approvals, revenue and payout summary
- Unified lead queue across Booking and RSVP
- Audit logs on approvals, payouts, rate changes, promotions, deactivations, document access
- Settings: roles, rates, assessment questions, notification templates, geofence radius, confirmation windows

Done when: every role sees only its scope, enforced server-side, and an authorisation test suite covers each role against each endpoint.

6.12 RSVP Guest Management

Cadence, per event. Every step configurable, none hard-coded.

When	Action	Channel
Upload	Guest list imported via CSV or entered manually	—
T-30	Automated invitation message to all guests	WhatsApp
T-28	Team manually calls all guests to confirm availability; response recorded	Phone
After calls	Event manager informs host which guests are unavailable so arrangements adjust	Report
T-25	Automated request for masked identity document	WhatsApp
Ongoing	Responders marked received; non-responders flagged for follow-up calls	CRM
T-15 / 10 / 7 / 5 / 3	Five automated reminders, triggered once document collection completes	WhatsApp
Event day	Transport mode requested — road, flight with PNR, train with PNR — surfaced to logistics	WhatsApp
T+2 / T+3	Feedback message to guests	WhatsApp

- Guest statuses: Confirmed · Pending · Not Coming · Follow-up Required · Document Required · Document Received



-
- At-a-glance document status view, Received versus Pending
 - Final RSVP summary report generated and shared with client or planner
 - All WhatsApp templates classified as UTILITY, never marketing — marketing rates are roughly seven times higher

Done when: the full cadence runs from a single uploaded CSV without manual scheduling, and the deletion job removes documents after the retention window.

7. Non-Functional Requirements

Area	Requirement
Performance	Every query on a collection above 100 rows uses an index. The PR must name the index it relies on
Data loading	Pagination, server-side filtering, debounced search. Never load a full collection
Concurrency	Slot claiming and payout writing must be safe under simultaneous load
Mobile	Freelancer screens are mobile-first and tested on a real mid-range Android device
Security	Server-side authorisation on every endpoint. Never trust an ID from a request body — derive identity from the session
Validation	Input validated at the API boundary, not only in the UI
Secrets	No secrets in the repository. No sensitive values in logs
Audit	Every approval, payout, rate change and document access written to AuditLog
Errors	Loading, empty and error states handled on every screen. No silent failures
Backups	Daily automated, plus a manual snapshot before each release

8. Delivery Plan

Days	Phase	Output
1-3	Foundation	Types, API contract, design system, CI, one complete vertical slice as the reference pattern
4-7	Identity and public site	Auth, RBAC, freelancer onboarding and assessment, public website
8-14	Core operations	Planner and client flows, venue database, events, positions, FCFS allocation
15-19	Field and finance	Attendance, ratings, earnings, payouts, quotations, invoices, expenses
20-22	RSVP and admin	Guest workflow, WhatsApp cadence, admin dashboard, reports
23-25	Integration	Cross-module integration, concurrency testing, security pass
26-27	UAT	Client review, defect fixes
28	Deployment	Production deploy, handover, training

Ownership: Dev A — identity and freelancer. Dev B — demand, events and attendance. Dev C — finance, RSVP and admin. Each owns their slice end to end: UI, API, data and tests. Nobody edits another zone; they request the change.

External blocker: payout provider merchant verification is outside our control and routinely takes one to two weeks. The application must be filed before day one, not during the sprint.

9. Known Traps

These three are where AI-generated code on this project is reliably wrong. Each one requires a human review and a passing test before merge.

9.1 The claim race

Every model will write read-count, check-space, then write-claim. It looks correct and it is not. Two simultaneous claims both read nine filled and both succeed. Eleven people arrive for ten slots and you discover it at the venue.

Required: a single atomic conditional update that only succeeds when a slot is genuinely free.

```
test: 20 concurrent claims on 10 slots → exactly 10 assigned
test: 500 concurrent claims on 10 slots → exactly 10 assigned
```

9.2 Payout double-pay

A retried or duplicated payout request that produces a second transfer is real money leaving a real account, and it is not recoverable by a code fix.

Required: idempotency key on every payout, enforced at the database level, not only in application logic.

```
test: same payout submitted twice → one transfer
test: payout cannot reach `paid` without a providerTxnId
test: batch interrupted mid-run and re-run → no duplicates
```

9.3 Missing indexes

Agents write working queries that scan whole collections. Invisible at fifty rows, fatal at five thousand — and it is what pushes the database off the cheap tier years early, breaking the cost model quoted to the client.

Required: every query on a collection above 100 rows uses an index, named in the PR description.

```
Minimum indexes: freelancerId, eventId, positionId, status,
                  claimedAt, location.city, month+freelancerId
```

10. Scope Boundary

The following are explicitly NOT in Phase 1. An agent must not build them, and must not add scaffolding for them beyond what the API-first architecture already provides.

- Native Android and iOS applications
- Continuous background geofencing with live exit alerts
- AI voice calling and automated guest conversations
- Advanced analytics, forecasting and AI-assisted insights
- Advanced vendor management
- Multi-language support

A new client request during the sprint goes on a list. It does not go into the sprint. The list becomes the Phase 1.1 conversation after launch.